The University of Manchester

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 6 Practical - Introduction to Integration (Solutions)

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Describe the concept of integration and explain its significance in calculus.

(2) Identify and interpret common notation used in integration.

(3) Describe the role of Ordinary Differential Equations (ODEs) in scientific modeling across various disciplines.

(4) Apply common laws of integration to solve theoretical and real-world problems.

Exercise 1: Common Laws of Integration

- Evaluate the following indefinite integrals:

$$\int 6x^5 - 18x^2 + 7\,dx$$

$$\int 10w^4 + 9w^3 + 7w\,dw$$

- You can verify your answer by differentiating and making sure that the result is the same as the integrand!

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

Exercise 1: Common Laws of Integration

- Evaluating the first integral:

$$\int 6x^5 - 18x^2 + 7\,dx = \frac{6}{6}x^6 - \frac{18}{3}x^3 + \frac{7}{1}x + d$$
$$= x^6 - 6x^3 + 7x + d$$

- Evaluating the second integral:

$$\int 10w^4 + 9w^3 + 7w\,dw = \frac{10}{5}w^5 - \frac{9}{4}w^4 + \frac{7}{2}w^2 + d$$
$$= 2w^5 - \frac{9}{4}w^4 + \frac{7}{2}w^2 + d$$

- Don't forget the constants in indefinite integrals!

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
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Exercise 2: Definite Integration

- A gas leaks from a container according to the following ordinary differential equation:

$$\frac{dG}{dt} = 390t^{0.4}$$

where t is time (in minutes) and G is in litres per minute.

- Calculate the volume of gas that has been released between 1 and 3 minutes.

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
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Solution 2: Definite Integration

- As we want to calculate the volume of gas that has been released between 1 and 3 minutes, we integrate $\frac{dG}{dt}$ between 1 and 3:

$$\begin{aligned}\int_1^3 390t^{0.4} &= 390\left[\frac{t^{1.4}}{1.4}\right]_1^3 \\ &= 390\left(\frac{3^{1.4}}{1.4} - \frac{1^{1.4}}{1.4}\right) \\ &= 390\,(3.325 - 0.714) = 1018.328\end{aligned}$$

- So approximately 1018.328 litres were lost between 1 and 3 minutes.

Derivative $\frac{dy}{dx}$	**Integral** y
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Exercise 3: Hydrostatic Equation

- In lectures, we aimed to solve the Hydrostatic Equation, which states that air density depends on pressure, P, and temperature, T, through the ideal gas law, $\rho = \frac{P}{R_\alpha T}$:

$$\frac{dP}{dz} = -\frac{g}{R_\alpha T} P$$

where g and R_α are constants.

- We used separation of variables to get the following integrals:

$$\int \frac{1}{P} dP = \int -\frac{g}{R_\alpha T} dz$$

- Complete the integration to show that:

$$P(z) = P_0 \exp\left(-\frac{g}{R_\alpha T} z\right)$$

is a solution to the Hydrostatic Equation. Here, P_0 is the pressure at surface level.

Solution 3: Hydrostatic Equation

- The integral $\int \frac{1}{x} dx$ is equal to $\ln x$ so we have

$$\begin{aligned} \ln P &= -\frac{g}{R_\alpha T} z + c \end{aligned}$$

and if we say at $t = 0$, $P = P_0$ we get that $c = \ln(P_0)$, we therefore obtain

$$\begin{aligned} P(z) &= \exp\left(-\frac{g}{R_\alpha T} z + \ln(P_0)\right) \\ &= P_0 \exp\left(-\frac{g}{R_\alpha T} z\right) \end{aligned}$$

as the solution to Hydrostatic Equation.

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